

WindOps Copilot - Turbine Report

Turbine ID: WTG-001 | Correlation ID: corr-7e88e09d8b8b
Report Generated: 2026-06-26T10:18:42.031667

TURBINE HEALTH STATUS

50.0% Health

Live Sensor Telemetry

Sensor Parameter	Value	Unit
Wind Speed	12.94	m/s
Rotor Speed	16.50	RPM
Blade Temperature	24.19	°C
Generator Temperature	73.80	°C
Vibration	0.28	mm/s
Power Output	2.08	MW
Operational Status	active	

Active Fault Alerts

Parameter	Value	Threshold	Severity
Vibration	0.28	0.25	CRITICAL
Generator Temp	73.80	65.00	WARNING

Diagnostic Root Cause Analysis

Diagnosed Cause:	Generator Bearing Degraded / Gearbox Defect
Confidence Level:	85.0%
Technical Analysis:	Vibrations are high (0.28 mm/s) accompanied by elevated generator temperatures (73.8°C). This indicates friction, misalignment, or bearing breakdown in the drive train/shaft.

Operational Action Recommendations

Recommended Action	Description	Risk	Production Loss	Confidence
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Emergency Controlled Shutdown	Trigger pitch feathers and rotor brake immediately due to critical vibration threshold breach (>0.25 mm/s). Protect structural integrity of the asset.	LOW	severe	100%
De-rate / Curtail Power Output	Reduce generator speed. (NOT RECOMMENDED: safety thresholds are violated, structural damage is likely if kept spinning).	HIGH	moderate	10%

Note: This report is generated by an autonomous multi-agent system. final actions must be reviewed and authorized by a human operator.